BUS13-4

Microeconomics

Choice under constraint, prices, and the limits of markets

Every actor chooses under constraint; prices coordinate those choices — and where they cannot, markets fail.

Albert School — 22 hours

Contents

Preface

This is the entry point to the Economics arc at Albert, and it assumes no economics before it. What it does assume is a willingness to reason carefully about choices — and a little comfort with linear equations, since most of our models reduce to solving two lines for where they cross.

The promise of the course is a single, portable habit of mind. An economist looks at any situation involving scarce resources and asks three questions in order: **what does this choice cost in terms of what it forecloses** (opportunity cost); **is one more unit worth it** (marginal reasoning); and **what do the incentives push people to do** (response to the margin). Almost everything else — supply and demand, elasticities, cost curves, monopoly, externalities — is this habit applied to a new setting, made precise with a diagram and an equation.

We teach each idea the way understanding actually forms: a concrete number first, the general principle second. A handful of small fictional markets — a coffee stand, a lighting firm, a town with a polluting factory — recur so that the abstractions always have something to point at. We also keep an honest running commentary on where the standard "rational actor" story breaks down, because the gap between how people are assumed to choose and how they actually choose is itself one of the subject's most useful lessons.

The course closes with a short bridge into how a whole economy is measured, so that **BUS23-4 Macroeconomics** can open directly on money, central banking and the multiplier without re-explaining what GDP is.

1 Foundations of economic reasoning

Economics begins with an uncomfortable fact: resources are **scarce**. There is not enough time, money, land or attention to do everything worth doing, so every choice to use a resource one way is simultaneously a choice **not** to use it every other way. The discipline is, at bottom, the study of how people, firms and societies cope with that constraint.

1.1 Opportunity cost: the cost that money does not show

> **Worked example — the free concert.** A friend gives you a free ticket to a concert tonight. The ticket cost you nothing. But going still has a cost: tonight you could instead have worked a €60 shift, or studied for tomorrow's exam. If the best thing you give up is the €60 shift, then attending the "free" concert costs you €60. The price tag and the cost are not the same number.

Definition 1 The **opportunity cost** of a choice is the value of the best alternative forgone in making it. It is the true economic cost of any decision, whether or not money changes hands. Whenever you read "cost" in this book, read it as opportunity cost unless stated otherwise.

Opportunity cost is why an economist is never impressed by the word "free." A free ticket, a free afternoon, a factory you already own — each has a cost equal to its most valuable alternative use.

1.2 Marginal reasoning: think in extra units, not totals

Big decisions are rarely all-or-nothing. The useful question is almost never "should this activity exist?" but "should I do a little **more** or a little **less** of it?" That is **marginal** thinking.

> **Worked example — how many coffees to brew.** A coffee stand sells cups for €3 each. The first cups of the morning are cheap to make; but to serve the 200th cup the owner must stay open an extra hour, pay overtime, and brew pots that go stale — say €4 of extra cost for that cup. The 200th cup brings in €3 of extra revenue and costs €4: it destroys €1 of value, even though the stand as a whole is wildly profitable. The right scale is found by pushing output to the point where the **next** cup's extra benefit just meets its extra cost — not by looking at the total.

Definition 2 **Marginal benefit** (MB) and **marginal cost** (MC) are the change in total benefit and total cost from one additional unit of an activity. **Marginal reasoning** says: expand an activity while $MB > MC$, stop when $MB = MC$, and contract it when $MB < MC$. The optimum is at the margin, not the average.

This single rule — $MB = MC$ — reappears, lightly disguised, as the consumer's tangency condition, the firm's $MC = MR$ rule, and the social optimum behind a Pigouvian tax. Learn it once here.

1.3 Sunk costs: the past is not a cost

> **Worked example — the half-eaten buffet.** You pay €25 for an all-you-can-eat buffet. After two plates you are full. A voice says: "eat more — you paid for it." But the €25 is gone whether you eat one plate or five. The only live question is whether a third plate's enjoyment beats its discomfort. The €25 should play **no** part in that decision.

Definition 3 A **sunk cost** is an expenditure already incurred that cannot be recovered. Sunk costs are **irrelevant** to current decisions: only costs and benefits that still change with the choice in front of you matter. Spending more because you have "already invested so much" is the **sunk-cost fallacy**.

Common pitfall The two most common foundational errors are mirror images. (1) Ignoring opportunity cost — treating a resource you already own (your time, an idle machine) as free. (2) Honouring sunk cost — letting money already spent justify spending more. The discipline is the same in both: count only the costs and benefits that **change with the decision you are making now**.

1.4 Incentives

Definition 4 An **incentive** is any feature of the environment — a reward or a penalty — that changes the marginal benefit or marginal cost of an action, and therefore changes behaviour. Because rational actors respond at the margin, even small, well-placed incentives can move behaviour a lot, and badly designed ones produce perverse results (a fine for late pickup at a daycare, framed as a price, can **increase** lateness).

1.5 The production possibility frontier

A society, like a person, faces opportunity cost. The **production possibility frontier** makes that trade-off visible for an economy producing two goods.

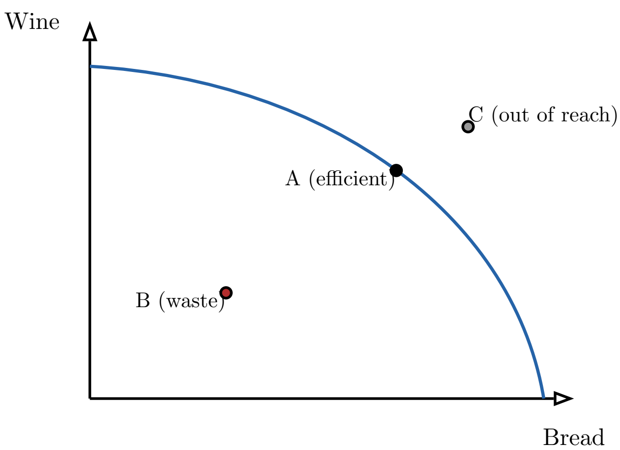

Figure 1: The production possibility frontier for an economy making bread and wine. Points on the curve (A) use all resources efficiently; points inside (B) waste resources; points outside (C) are unattainable with current resources and technology. The frontier bows outward because resources are not equally good at both tasks, so each extra loaf of bread costs ever more wine — its slope is the opportunity cost of bread in terms of wine.

Definition 5 The **production possibility frontier** (PPF) is the set of maximum output combinations attainable from fixed resources and technology. Its (negative) slope at any point is the opportunity cost of one more unit of the good on the horizontal axis, measured in units of the good on the vertical axis. Its outward bow reflects **increasing** opportunity cost.

1.6 Positive versus normative

Worked example — two sentences about a minimum wage. “Raising the minimum wage by 10% reduces teenage employment by about 1%.” — a claim about what **is**, testable against data. “The government ought to raise the minimum wage.” — a claim about what **should be**, resting on values. The first can be checked; the second cannot be settled by evidence alone, only informed by it.

Definition 6 A **positive** statement describes what is or predicts what will be; it can in principle be tested against evidence. A **normative** statement prescribes what ought to be; it embeds a value judgement. Good economic argument keeps the two separate — most policy disputes mix a positive disagreement about effects with a normative disagreement about goals, and progress starts by telling them apart.

2 Bounded rationality and behavioral realism

Almost every model in this book assumes a particular kind of actor.

Definition 7 The **rationality hypothesis** assumes actors have consistent, well-ordered preferences and choose, given their constraints and information, the option that best satisfies them. It is the engine of the rational-choice predictions used throughout the course.

The hypothesis is a powerful simplification, and often a good approximation. But decades of evidence show systematic, **predictable** departures from it. Knowing them is not a rejection of the toolkit — it is knowing when the toolkit's predictions need a correction, and it is essential for **designing incentives** that work on real people.

Definition 8 **Bounded rationality**: actors have limited information, attention and computational power, so they **satisfice** — settle for a good-enough option — rather than optimise. (Herbert Simon's term.)

Worked example — framing the surgery. Told “this operation has a 90% survival rate,” most patients consent. Told the logically identical “this operation has a 10% mortality rate,” far fewer do. The facts are the same; only the description changed.

Definition 9 **Framing**: choices depend on how options are described, not only on their substance. A gain frame and a loss frame for the same outcome can flip a decision.

Definition 10 **Representativeness**: judging probability by similarity to a stereotype rather than by base rates — e.g. assuming a quiet, bookish person is “more likely” a librarian than a salesperson, ignoring that salespeople vastly outnumber librarians.

Definition 11 **Loss aversion**: a loss hurts more than an equal gain pleases — empirically, by roughly a factor of two. It explains why people cling to losing positions and demand more to give up something than they would pay to get it.

Definition 12 **Present bias**: immediate payoffs are overweighted relative to future ones, so plans made for "tomorrow" are reversed when tomorrow becomes today — the source of procrastination and under-saving.

Common pitfall These findings belong to **incentive design**, and that is a different use from marketing's consumer psychology. Marketing uses framing and the rest to **persuade** — to sell more. Incentive design uses them to **align behaviour** with a goal: auto-enrolling employees in a pension (defaults beat exhortation because of present bias), or paying a bonus as a sum that can be **lost** rather than gained (loss aversion). Same findings; opposite intent. Do not conflate the two — this course is about the second.

3 Supply, demand, and market equilibrium

A competitive market is described by two relationships between the price p of a good and the quantity q of it.

Worked example — the coffee market in numbers. Suppose buyers in a town demand $q_d = 100 - 10p$ cups per hour and sellers supply $q_s = 20 + 10p$ cups per hour, with p in euros. At €3 buyers want $100 - 30 = 70$ cups but sellers offer $20 + 30 = 50$: a shortage of 20 pushes the price up. At €5 buyers want 50 and sellers offer 70: a surplus pushes the price down. Somewhere between lies the price that clears the market.

Definition 13 A **demand curve** gives the quantity buyers wish to buy at each price, normally downward-sloping: $q_d = a - bp$ with $a, b > 0$. A **supply curve** gives the quantity sellers wish to sell at each price, normally upward-sloping: $q_s = c + dp$ with $d > 0$.

Definition 14 **Market equilibrium** is the price p^* at which quantity demanded equals quantity supplied, found by solving $a - bp = c + dp$:

$$p^* = \frac{a - c}{b + d}, \quad q^* = a - bp^*.$$

At p^* there is neither shortage nor surplus, so nothing pushes the price to move. For the coffee market: $100 - 10p = 20 + 10p \Rightarrow p^* = 4$, $q^* = 60$.

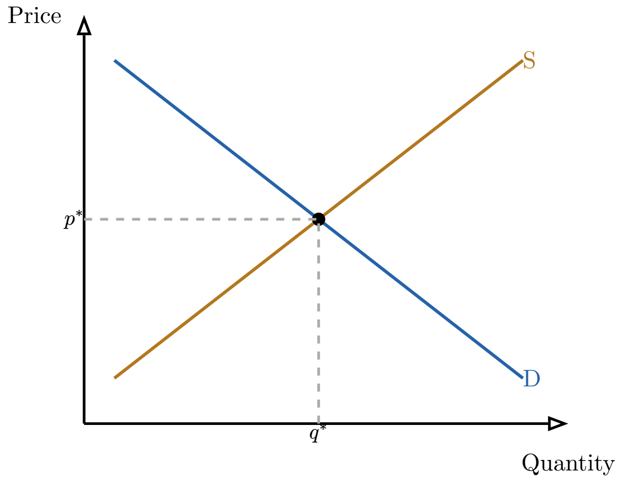

Figure 2: Equilibrium sits where supply meets demand. Above p^* supply exceeds demand (a surplus that pushes price down); below it demand exceeds supply (a shortage that pushes price up). Only at p^* is there no pressure to move.

Definition 15 A **demand shock** shifts the whole demand curve (a change in the intercept a); a **supply shock** shifts the supply curve (a change in c). Re-solve $q_d = q_s$ with the new intercept to find the new equilibrium. The qualitative rules: an increase in demand raises both p^* and q^*; an increase in supply lowers p^* and raises q^*.

Worked example — a frost hits the coffee crop. A frost cuts supply to $q_s = 0 + 10p$ (the intercept c falls from 20 to 0). Re-solving: $100 - 10p = 10p \Rightarrow p^* = 5$, $q^* = 50$. The supply shock raised the price (€4 → €5) and cut the quantity (60 → 50), exactly as the rule predicts.

Common pitfall Keep "a change in quantity demanded" (a **move along** a fixed demand curve, caused by the good's own price) distinct from "a change in demand" (a **shift** of the whole curve, caused by income, tastes, or other goods' prices). Saying "demand rose because the price fell" confuses the two and will mislead every shock analysis you attempt.

4 Elasticities

Slopes depend on units — cups per euro is a different number from litres per cent. **Elasticities** strip out units by working in percentages, so they compare across markets and are the natural language for "how responsive is one thing to another?"

Worked example — two price cuts, two outcomes. A pharmacy cuts the price of a life-saving drug by 10% and sells only 2% more — buyers need it regardless. A café cuts the price of one brand of soda by 10% and sells 30% more — buyers happily switch from rivals. Same 10% price cut; wildly different responsiveness. Elasticity is the number that captures the difference.

Definition 16 The **price elasticity of demand** is the percentage change in quantity demanded per percentage change in price:

$$\varepsilon_p = \frac{\%\Delta q}{\%\Delta p} = \frac{dq}{dp} \cdot \frac{p}{q}.$$

It is negative for normal downward-sloping demand; we compare its magnitude. Demand is **elastic** if $|\varepsilon_p| > 1$ (the pharmacy is **inelastic**, $|\varepsilon_p| = 0.2$; the soda is **elastic**, $|\varepsilon_p| = 3$), **inelastic** if $|\varepsilon_p| < 1$, and **unit-elastic** if $|\varepsilon_p| = 1$.

Definition 17 The **income elasticity** $\varepsilon_I = \frac{dq}{dI} \cdot \frac{I}{q}$ measures responsiveness to income I. A good is **normal** if $\varepsilon_I > 0$ and **inferior** if $\varepsilon_I < 0$ (bus rides may fall as income rises). The **cross-price elasticity** $\varepsilon_{XY} = \frac{dq_X}{dp_Y} \cdot \frac{p_Y}{q_X}$ measures how good X's demand responds to good Y's price: positive for **substitutes** (butter and margarine), negative for **complements** (printers and ink).

4.1 Elasticity and total revenue

The most useful business payoff of elasticity is that it tells you, before you act, whether a price change will raise or lower revenue $R = pq$.

Worked example — should the café raise prices?. For the elastic soda ($|\varepsilon_p| = 3$), a 10% price rise loses about 30% of sales — revenue falls. For the inelastic drug ($|\varepsilon_p| = 0.2$), a 10% rise loses only 2% of sales — revenue rises. The rule: raise price on inelastic goods, cut price on elastic ones.

Definition 18 Because revenue is price times quantity, a price rise has two opposing effects — more euros per unit, fewer units. Which wins is decided by elasticity:

- inelastic demand ($|\varepsilon_p| < 1$): price and revenue move **together**;
- elastic demand ($|\varepsilon_p| > 1$): price and revenue move in **opposite** directions;
- unit-elastic ($|\varepsilon_p| = 1$): revenue is unchanged — it is at its maximum.

Common pitfall Elasticity is not the slope. A straight-line demand curve has a **constant** slope but a **changing** elasticity: demand is elastic at high prices (near the top) and inelastic at low prices (near the bottom), passing through unit-elastic at the midpoint. So “is this market elastic?” has no answer without saying **at what price**.

5 Welfare: surplus and deadweight loss

Markets are not just efficient or inefficient in the abstract; we can **measure** the gains they create and the losses when they are distorted, in the same euros.

Definition 19 **Consumer surplus** (CS) is the area between the demand curve and the price, up to the quantity traded: total willingness to pay minus what buyers actually paid. **Producer surplus** (PS) is the area between the price and the supply curve: revenue minus sellers’ minimum acceptable receipts. **Total surplus** = CS + PS is the market’s total gains from trade, and is maximised at the competitive equilibrium.

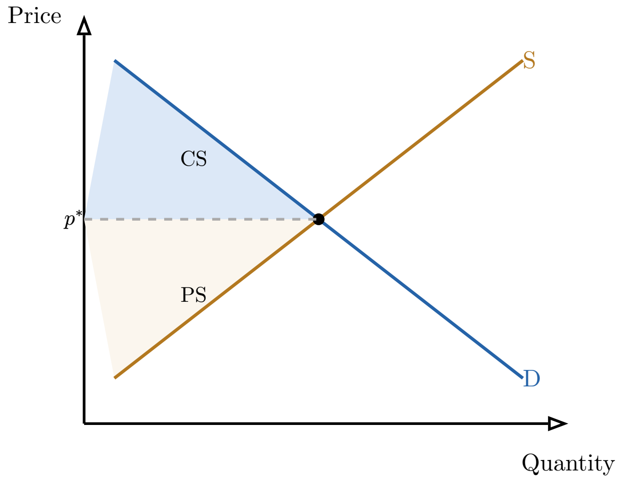

Figure 3: Consumer surplus (upper, blue) is the value buyers get above the price they pay; producer surplus (lower, sand) is what sellers get above their minimum. At the competitive equilibrium the two together — the whole triangle between supply and demand up to q^* — are as large as they can be.

> **Definition 20** **Deadweight loss** (DWL) is the reduction in total surplus when the traded quantity departs from the competitive level — caused by a tax, by market power, or by any other wedge. It is value that **no one** captures: trades that would have benefited both buyer and seller simply do not happen.

6 Tax incidence

Who really pays a tax is rarely who the law names. A tax "on sellers" can land mostly on buyers, and vice versa — the answer is set by elasticities, not by the statute.

> **Worked example — a €2 tax on each cup of coffee.** Return to $q_d = 100 - 10p_c$ and $q_s = 20 + 10p_s$, equilibrium €4 / 60 cups. Levy a €2 per-cup tax, so buyers pay p_c and sellers keep $p_s = p_c - 2$. Setting $q_d = q_s$: $100 - 10p_c = 20 + 10(p_c - 2) \Rightarrow 100 - 10p_c = 0 + 10p_c \Rightarrow p_c = 5$, $p_s = 3$, $q = 50$. Buyers pay €1 more (€4 → €5), sellers receive €1 less (€4 → €3): here the €2 splits evenly, because demand and supply are equally elastic.

> **Definition 21** A **per-unit tax** t drives a wedge $p_c - p_s = t$ between the price buyers pay and the price sellers receive. With linear demand slope b and supply slope d (magnitudes), the share of the tax borne by **consumers** is
>
> $$\frac{d}{b+d},$$
>
> and by producers $b/(b + d)$. The burden falls more heavily on the **less elastic** side — the side less able to walk away.

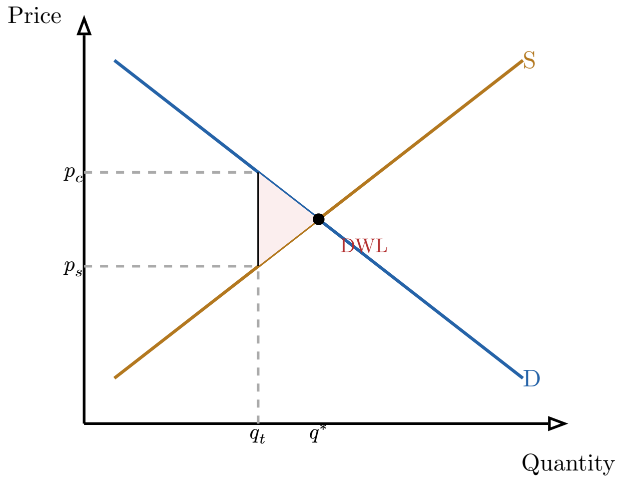

Figure 4: A per-unit tax opens a wedge between the price buyers pay (p_c) and the price sellers receive (p_s), shrinking the traded quantity from q^* to q_t. The rectangle between p_c and p_s up to q_t is government revenue; the red triangle to the right is the deadweight loss — mutually beneficial trades the tax destroys.

The tax's total cost to the market splits into three parts: revenue $t \times q_t$ that the government collects, surplus **transferred** from buyers and sellers to the government, and the **deadweight loss** — the triangle of trades that no longer happen, which benefits no one.

Common pitfall The **statutory** side of a tax (who writes the cheque) tells you nothing about the **economic** side (whose surplus actually falls). Politicians announce "a tax on corporations, not households," but if demand is inelastic the household pays it through a higher price all the same. Always compute incidence from elasticities, never from the label on the law.

7 Consumer theory

We now look **inside** the demand curve: where does a consumer's willingness to pay come from? The answer is a simple picture — what you want (indifference curves) laid against what you can afford (the budget line).

Definition 22 A **utility function** $u(x, y)$ assigns a number to each bundle of two goods, representing the consumer's preference ranking. An **indifference curve** joins all bundles giving equal utility; the consumer is indifferent along it. Its slope (in magnitude) is the **marginal rate of substitution** (MRS) — how much y the consumer will trade for one more x while staying equally happy. Indifference curves are downward-sloping and bowed toward the origin (diminishing MRS: the more x you already have, the less y you will give up for more of it).

Definition 23 The **budget constraint** is the set of affordable bundles, $p_x x + p_y y = I$ for income I. It is a straight line with slope $-p_x/p_y$ — the rate at which the **market** lets you swap y for x.

Definition 24 The consumer's **optimal choice** is the affordable bundle on the highest reachable indifference curve. There the indifference curve is **tangent** to the budget line, so the rate at which the consumer is **willing** to trade equals the rate at which the market **lets** them trade:

$$\text{MRS} = \frac{p_x}{p_y}.$$

(This is the marginal rule MB = MC again, one good against the other.)

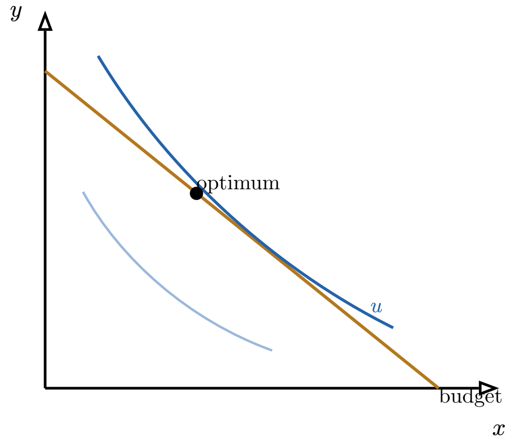

Figure 5: The consumer climbs to the highest indifference curve the budget line allows. The best affordable bundle is where an indifference curve just touches the budget line — the tangency $\text{MRS} = p_x/p_y$. A higher curve would be preferred but is unaffordable; a lower one is affordable but worse.

7.1 Decomposing a price change

When a price falls, the consumer buys more for two distinct reasons, and separating them is one of the most useful moves in applied microeconomics.

> **Worked example — the price of coffee falls.** Coffee gets cheaper. You buy more for two reasons. First, coffee is now cheap **relative to tea**, so you swap some tea for coffee even at the same standard of living — the **substitution** effect. Second, your same income now stretches further; you are effectively richer, and buy more coffee on that account too — the **income** effect.

Definition 25 The **substitution effect** is the change in quantity demanded due to the change in **relative** prices, holding utility (real living standard) constant — always toward the now-cheaper good. The **income effect** is the change due to the altered **purchasing power** a price change creates. The observed total response is their sum. For a normal good the two reinforce; for an inferior good they oppose.

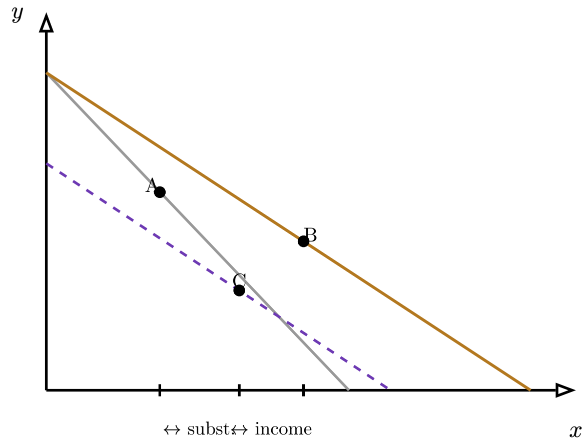

Figure 6: Decomposing a fall in the price of x. The move from the old optimum A to the new optimum B splits into a substitution step A→C (slide along the original satisfaction level to the new relative prices, via the dashed compensated budget) and an income step C→B (the parallel shift out to the true new budget). For a normal good both steps raise x.

The business reading: the substitution effect is customers switching toward your product when it becomes relatively cheaper; the income effect is the extra demand that comes simply because a price cut leaves buyers with more spending power overall.

8 Firm theory: production and costs

We now cross to the supply side and open up the firm. Where does the upward supply curve come from? From the firm's costs, and those come from its technology.

Definition 26 A **production function** $q = f(L, K)$ gives output from inputs — typically labour L and capital K. The **marginal product** of an input is the extra output from one more unit of it, holding others fixed.

Definition 27 **Diminishing marginal returns**: holding capital fixed, each extra worker eventually adds **less** output than the last — a kitchen with one oven gets crowded. This is what ultimately makes marginal cost rise, and so makes supply curves slope up.

8.1 From production to costs

Definition 28 **Fixed cost** (FC) does not vary with output (rent, a leased machine); **variable cost** (VC) does (materials, hourly labour). Total cost $\mathrm{TC} = \mathrm{FC} + \mathrm{VC}$. Per-unit measures: **average total cost** $\mathrm{ATC} = \mathrm{TC}/q$, **average variable cost** $\mathrm{AVC} = \mathrm{VC}/q$, and **marginal cost** $\mathrm{MC} = (d\ \mathrm{TC})/(dq)$, the cost of one more unit. Because of diminishing returns, MC eventually rises; and because fixed cost is spread over more units, ATC first falls then rises, giving the characteristic U-shape.

Worked example — a small bakery's costs. Daily fixed cost (oven lease, rent) is €120. Variable cost rises with loaves: 100 loaves cost €100 of flour and labour, 200 loaves €260 (overtime, a second shift). Then at 200 loaves $\mathrm{ATC} = (120 + 260)/200 = €1.90$ and $\mathrm{AVC} = 260/200 = €1.30$. The €0.60 gap is fixed cost spread thin — and it shrinks as output grows, which is why ATC keeps falling while output is low.

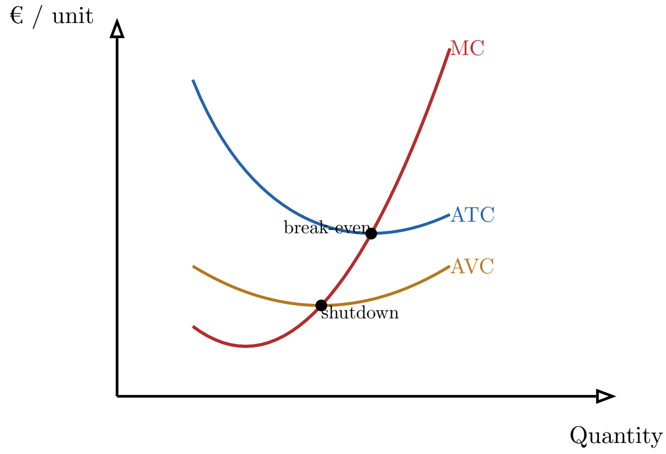

Figure 7: The cost curves. Marginal cost (red) cuts both average curves at their lowest points. Where MC meets minimum ATC is the **break-even** price (zero economic profit); where MC meets minimum AVC is the **shutdown** price — below it the firm cannot even cover its variable cost and stops producing in the short run.

Definition 29 The **break-even point** is the output where $p = \text{ATC}$ at its minimum: economic profit is zero. The **shutdown point** is the output where $p = \text{AVC}$ at its minimum: below this price the firm loses **more** by operating than by stopping, so in the short run it produces nothing. Between the two it keeps operating at a loss — because it at least covers variable cost and part of the unavoidable fixed cost.

Common pitfall Shutdown is a **short-run** decision governed by AVC, not ATC. A firm losing money (price below ATC) should still produce as long as price clears AVC, because the fixed cost is sunk in the short run — there it is again, the sunk-cost principle. Comparing price to ATC to decide whether to **operate today** is the classic error.

9 Profit maximization

Definition 30 **Marginal revenue** (MR) is the extra revenue from selling one more unit. A firm maximises profit at the output where

$$\text{MC} = \text{MR},$$

provided price covers AVC. Below this output an extra unit adds more revenue than cost (make it); above it, more cost than revenue (don't).

Under **perfect competition** the firm is a **price-taker**: it can sell all it wants at the going price and none above it, so $\text{MR} = p$ and the rule becomes $\text{MC} = p$. The marginal-cost curve (above AVC) **is** the firm's supply curve.

Definition 31 The **zero-profit theorem**: under perfect competition with free entry and exit, economic profit tends to zero in the long run. Positive profit attracts new firms, raising supply and lowering price; losses drive firms out, cutting supply and raising price; the process stops only when $p = \text{ATC}_{\text{min}}$ and profit is zero.

Common pitfall "Zero economic profit" does not mean the owner earns nothing. Economic cost already includes the **opportunity cost** of the owner's capital and effort — a normal return. Zero economic profit means the firm earns exactly that normal return and no more: just enough to stay, not enough to attract a crowd.

10 Perfect and monopolistic competition

Definition 32 **Perfect competition** assumes many price-taking firms, a homogeneous product, and free entry and exit. Each firm sets $p = \text{MC}$; in the long run entry drives the market to $p = \text{ATC}_{\text{min}}$ with zero economic profit. It is the efficiency benchmark every other market structure is judged against.

Definition 33 **Monopolistic competition** keeps the many firms and free entry but makes products **differentiated** — each firm sells something slightly distinct (cafés, hair salons, brands of cereal) and so faces its own downward-sloping demand curve. In the short run a firm can earn profit by setting $\text{MC} = \text{MR}$; entry of close substitutes erodes it until long-run profit is again zero — but now at an output **below** minimum-ATC scale, leaving some excess capacity as the price of variety.

11 Monopoly and price discrimination

Worked example — the only pharmacy in town. A monopolist faces the **whole** market demand $p = 20 - q$ and has constant marginal cost $\text{MC} = 4$. Selling one more unit means cutting the price on **all** units, so marginal revenue falls twice as fast as price: $\text{MR} = 20 - 2q$. Setting $\text{MR} = \text{MC}$: $20 - 2q = 4 \Rightarrow q_m = 8$, and the demand curve gives $p_m = 20 - 8 = 12$. A competitive industry with the same costs would set $p = \text{MC} = 4$ and sell $q = 16$ — so the monopoly halves output and triples price.

Definition 34 A **monopoly** is the sole seller. Facing the entire downward-sloping demand curve, it produces where $\text{MC} = \text{MR}$ and charges the highest price that demand allows at that quantity. With linear demand $p = a - bq$, marginal revenue is $\text{MR} = a - 2bq$. Output is below, and price above, the competitive level, creating a **deadweight loss** — the surplus on the units between q_m and the competitive q that are never traded.

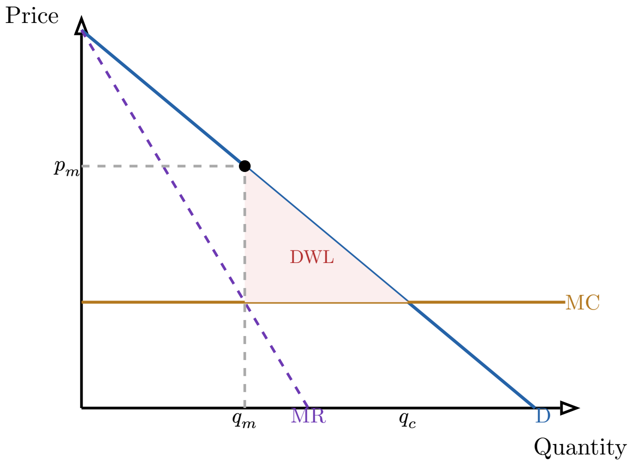

Figure 8: The monopoly outcome. The firm sets quantity where marginal revenue (dashed, twice as steep as demand) meets marginal cost, then prices up on the demand curve at p_m. Compared with the competitive quantity q_c (where demand meets MC), output is restricted, the price is higher, and the red triangle of lost mutually-beneficial trades is the deadweight loss.

Definition 35 **Price discrimination** charges different prices for the same good to capture surplus that a single price leaves on the table.

- **First-degree** (perfect): each unit sold at the buyer's exact maximum willingness to pay — e.g. a perfectly tailored negotiation. Captures **all** surplus.
- **Second-degree**: price varies with the quantity or version chosen, and buyers **self-select** — bulk discounts, airline fare classes, "good/better/best" tiers.
- **Third-degree**: different prices to identifiable **groups** with different elasticities — student and senior discounts, regional pricing. Charge the less-elastic group more.

All three require some market power and a way to prevent **resale** from the cheap segment to the dear one.

12 Oligopoly and strategic interaction

In an **oligopoly** a few firms share the market, and each one's best move depends on what the others do. This interdependence is the realm of **game theory**.

Definition 36 A **payoff matrix** lists the outcomes for each combination of players' strategies. A **Nash equilibrium** is a combination in which no player can do better by changing strategy **alone** — each is playing a best response to the others. To find it in a 2×2 game, mark each player's best response to each action of the rival; a cell where both are best responses is a Nash equilibrium.

Worked example — two firms choosing price — the prisoner's dilemma. Firms A and B each choose "High" or "Low" price. Payoffs are profits (A, B):

	B: High	**B: Low**
A: High	(8, 8)	(2, 10)
A: Low	(10, 2)	(4, 4)

Whatever B does, A earns more by choosing Low ($10 > 8$ if B High; $4 > 2$ if B Low), so Low is A's **dominant** strategy — and by symmetry B's too. Both play Low and earn (4, 4). Yet both would be better off at (8, 8) had they kept prices High. The unique Nash equilibrium (Low, Low) is **worse for both** than the cooperative outcome.

Definition 37 The **prisoner's dilemma** is any game in which each player has a dominant strategy to defect, yet mutual defection is worse for everyone than mutual cooperation. It is the canonical model of why cartels are unstable, why price wars break out, and why individually rational choices can be collectively self-defeating.

Common pitfall A Nash equilibrium is not the outcome the players would **jointly prefer**, and not necessarily efficient — the prisoner's dilemma proves both. "Equilibrium" means only "no one can gain by deviating alone," not "best for the group." Expecting Nash play to be socially optimal is a standard misreading.

13 Market failures

The competitive model promises efficiency, but only under conditions that real markets sometimes violate. Where they do, even free, voluntary trade leaves value on the table — a **market failure**. Two great families are externalities and asymmetric information.

13.1 Externalities and Pigouvian taxes

Worked example — a factory on the river. A factory's marginal **private** cost of output is $\mathrm{MC}_p = 2 + q$. But each unit also dumps €3 of pollution on downstream households — a cost the factory does not pay. The marginal **social** cost is $\mathrm{MC}_s = \mathrm{MC}_p + 3$. The factory produces where price meets **its** cost, so it overproduces relative to what is best for society as a whole.

Definition 38 An **externality** is a cost (negative) or benefit (positive) of a transaction that falls on a third party and is not reflected in the price. A negative externality (pollution) leads to **over**-production; a positive one (vaccination, education) to **under**-production — in both cases because the decision-maker ignores part of the true effect.

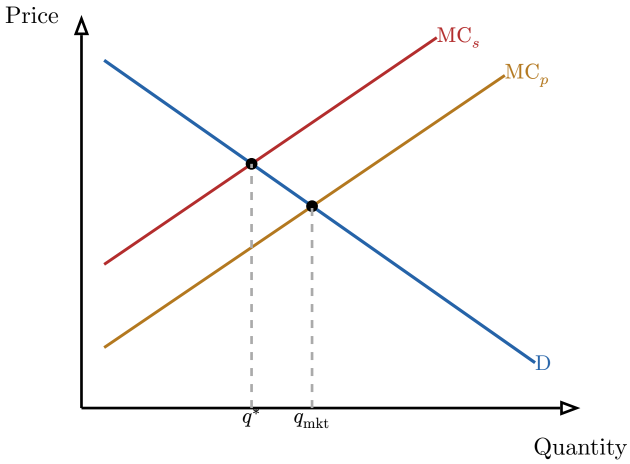

Figure 9: A negative externality. Firms supply along private marginal cost MC_p and the market settles at q_{mkt}, but the true cost to society is MC_s, higher by the external harm. The socially optimal quantity q^* is smaller. A Pigouvian tax equal to that vertical gap lifts MC_p onto MC_s and moves the market to q^*.

Definition 39 A **Pigouvian tax** is a per-unit tax set equal to the **marginal external cost at the socially optimal quantity**. By raising the firm's private marginal cost to the social marginal cost, it makes the decision-maker face the full cost of its action — it **internalises** the externality — and restores the efficient quantity. In the factory example the Pigouvian tax is €3 per unit, exactly the external harm.

13.2 Asymmetric information

Definition 40 **Asymmetric information** exists when one party to a transaction knows more than the other. It causes failure in two timings: **adverse selection** arises **before** the deal, when one side's hidden **type** (quality, risk) determines who chooses to trade; **moral hazard** arises **after**, when one side's hidden **action** changes once it is shielded from the consequences.

Worked example — Akerlof's market for lemons. Used cars are "peaches" (good, worth €10k) or "lemons" (bad, worth €4k), and only the seller knows which. A buyer, unable to tell them apart, will pay only the **average**, say €7k. But no owner of a €10k peach sells for €7k — so peaches withdraw, the cars left are mostly lemons, the fair average price falls, more good cars withdraw, and the market **unravels** toward only lemons. Quality collapses precisely because information is hidden.

Definition 41 Akerlof's **market for lemons** is the canonical model of **adverse selection**: when buyers cannot observe quality, hidden bad types drive out good ones, and trade that both sides would value at full information fails to happen. **Moral hazard** is its after-the-fact twin: an insured driver takes more risks, a worker on a fixed salary shirks — the hidden action shifts once its cost falls on someone else. Remedies (warranties, signalling, deductibles, monitoring) all work by restoring information or re-aligning incentives.

14 Bridge to macro measurement

The course closes by zooming out from single markets to the whole economy — not to do macroeconomics, but to define its central yardstick, so that **BUS23-4** can assume it.

Definition 42 **Gross domestic product** (GDP) is the market value of all **final** goods and services produced within a country in a given period. "Final" matters: counting both the flour and the bread made from it would **double-count** the flour, so GDP counts only final products — or, equivalently, the **value added** at each stage.

Definition 43 GDP can be measured three equivalent ways — they must agree because every euro of output is someone's expenditure and someone's income:

- **Production**: sum of value added across all producers.
- **Expenditure**: $\text{GDP} = C + I + G + (X - M)$ — consumption, investment, government spending, plus net exports (exports minus imports).
- **Income**: sum of incomes earned in production — wages, profits, rents, and taxes net of subsidies.

Common pitfall GDP measures **recorded market activity**, not welfare, and the gaps are large. It omits unpaid household and volunteer work and the informal economy; it ignores the **distribution** of income (it can rise while most people stagnate); it treats environmental degradation and resource depletion as free; and it credits some spending that merely repairs harm (cleaning up a disaster) as if it were genuine gain. A rising GDP is necessary information, but never the whole story of how a society is doing — the caveat **BUS23-4** will lean on from its first page.

The thread of this book. One habit — weigh marginal benefit against marginal cost, counting opportunity cost and ignoring sunk cost — generates everything here. It is the consumer's tangency, the firm's $\text{MC} = \text{MR}$, and the social optimum behind a Pigouvian tax. Prices coordinate these marginal choices and, in competition, do so efficiently; surplus and deadweight loss let us measure by how much. Where the conditions break — market power, externalities, hidden information, or the predictable ways real people depart from the rational ideal — markets fail in describable, often correctable ways. And GDP, for all its blind spots, is where the measurement of the whole economy begins.

Exercises

1. For a real resource-allocation decision of your choice, write one positive and one normative claim about it, identify the opportunity cost of the option chosen, and name a sunk cost that should play no role — explaining why.

2. Given linear demand $q_d = a - bp$ and supply $q_s = c + dp$, solve for equilibrium price and quantity. Then impose a stated demand or supply shock, re-solve, and compute the resulting consumer surplus, producer surplus, and any deadweight loss.

3. For a given market, compute the price, income, and cross-price elasticities of demand at a stated point, classify the good (elastic/inelastic, normal/inferior, substitute/complement), and predict whether a proposed price change raises or lowers total revenue.

4. For a per-unit tax t in a linear market, compute the price buyers pay, the price sellers receive, the new quantity, government revenue, the change in consumer and producer surplus, the deadweight loss, and the share of the burden borne by each side — and explain the split from the elasticities.
5. Given a utility function and a budget constraint, locate the consumer's optimal bundle using the tangency condition, then decompose the response to a stated price change into substitution and income effects, with a business interpretation of each.
6. State the rationality hypothesis, then describe bounded rationality, framing, loss aversion, and present bias, giving for each one concrete implication for incentive design that is distinct from a marketing application of the same finding.
7. From a table of production data, compute fixed, variable, marginal, average total, and average variable costs, and identify the firm's break-even and shutdown prices — justifying why shutdown is governed by AVC, not ATC.
8. Apply $\text{MC} = \text{MR}$ numerically to find a firm's profit-maximizing output under perfect competition, and explain, via free entry and exit, why economic profit tends to zero in the long run while the owner still earns a normal return.
9. For a linear-demand monopoly with constant marginal cost, compute output, price, and deadweight loss against the competitive benchmark; design a first-, a second-, and a third-degree price-discrimination scheme for it; and find the Nash equilibrium of a 2×2 pricing game between two such firms.
10. Diagnose a negative externality and compute the Pigouvian tax that internalizes it; then identify one adverse-selection situation (via Akerlof's market for lemons) and one moral-hazard situation, proposing a remedy for each.
11. Define GDP, work a small numerical example showing that the production, expenditure, and income approaches agree, and identify at least two significant phenomena GDP fails to capture, explaining why each matters.